

Assignment 3

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Necessary Packages

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(e1071)
```

```
##
```

```
## Attaching package: 'e1071'
```

```
## The following object is masked from 'package:ggplot2':
```

```
##
```

```
## element
```

```
set.seed(123)
```

Load Data

```
UniversalBank<-read.csv("~/Desktop/R documents/Assignment files/FML_Assignment2/UniversalBank.csv")
```

```
head(UniversalBank)
```

```
##   ID Age Experience Income ZIP.Code Family CCAvg Education Mortgage
## 1  1  25         1     49   91107      4   1.6         1         0
## 2  2  45        19     34   90089      3   1.5         1         0
## 3  3  39        15     11   94720      1   1.0         1         0
## 4  4  35         9    100   94112      1   2.7         2         0
## 5  5  35         8     45   91330      4   1.0         2         0
## 6  6  37        13     29   92121      4   0.4         2        155
##   Personal.Loan Securities.Account CD.Account Online CreditCard
## 1             0                 1           0         0         0
## 2             0                 1           0         0         0
## 3             0                 0           0         0         0
## 4             0                 0           0         0         0
## 5             0                 0           0         0         1
## 6             0                 0           0         1         0
```

Summary of Data

```
summary(UniversalBank)
```

```
##           ID           Age           Experience           Income           ZIP.Code
## Min.      : 1   Min.      :23.00   Min.      : -3.0   Min.      : 8.00   Min.      : 9307
## 1st Qu.:1251   1st Qu.:35.00   1st Qu.:10.0   1st Qu.: 39.00   1st Qu.:91911
## Median :2500   Median :45.00   Median :20.0   Median : 64.00   Median :93437
```

```
## Mean :2500 Mean :45.34 Mean :20.1 Mean : 73.77 Mean :93153
## 3rd Qu.:3750 3rd Qu.:55.00 3rd Qu.:30.0 3rd Qu.: 98.00 3rd Qu.:94608
## Max. :5000 Max. :67.00 Max. :43.0 Max. :224.00 Max. :96651
## Family CCAvg Education Mortgage
## Min. :1.000 Min. : 0.000 Min. :1.000 Min. : 0.0
## 1st Qu.:1.000 1st Qu.: 0.700 1st Qu.:1.000 1st Qu.: 0.0
## Median :2.000 Median : 1.500 Median :2.000 Median : 0.0
## Mean :2.396 Mean : 1.938 Mean :1.881 Mean : 56.5
## 3rd Qu.:3.000 3rd Qu.: 2.500 3rd Qu.:3.000 3rd Qu.:101.0
## Max. :4.000 Max. :10.000 Max. :3.000 Max. :635.0
## Personal.Loan Securities.Account CD.Account Online
## Min. :0.000 Min. :0.0000 Min. :0.0000 Min. :0.0000
## 1st Qu.:0.000 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:0.0000
## Median :0.000 Median :0.0000 Median :0.0000 Median :1.0000
## Mean :0.096 Mean :0.1044 Mean :0.0604 Mean :0.5968
## 3rd Qu.:0.000 3rd Qu.:0.0000 3rd Qu.:0.0000 3rd Qu.:1.0000
## Max. :1.000 Max. :1.0000 Max. :1.0000 Max. :1.0000
## CreditCard
## Min. :0.000
## 1st Qu.:0.000
## Median :0.000
## Mean :0.294
## 3rd Qu.:1.000
## Max. :1.000
```

Partitioning the data into training (60%) and validation (40%) sets

```
indextrain<- createDataPartition(UniversalBank$Personal.Loan, p=0.6, list=FALSE)

training<- UniversalBank[indextrain, ]

validation<- UniversalBank[-indextrain, ]
```

Create a pivot table

```
pivot_table <- table(CreditCard=training$CreditCard, PersonalLoan=training$Personal.Loan, Online=training$Online)

pivot_table
```

```
## , , Online = 0
##
##      PersonalLoan
## CreditCard    0    1
##      0  785   65
##      1  317   34
##
## , , Online = 1
##
##      PersonalLoan
## CreditCard    0    1
##      0 1145  122
##      1  475   57
```

Calculating probability

```
loan_accepted_online_CC <- pivot_table["1","1","1"]
total_CC_online <- sum(pivot_table["1", ,"1" ])
```

```
probability_loanaccepted <- loan_accepted_online_CC/total_CC_online
probability_loanaccepted
```

```
## [1] 0.1071429
```

Loan (rows) as a function of Online (columns) : Pivot table

```
loan_online_pivot <- table(PersonalLoan = training$Personal.Loan,
                           Online = training$Online)
loan_online_pivot
```

```
##           Online
## PersonalLoan  0    1
##           0 1102 1620
##           1   99  179
```

Loan (rows) as a function of Credit card (columns) : Pivot table

```
loan_CC_pivot<- table(PersonalLoan = training$Personal.Loan,
                      CreditCard = training$CreditCard)
loan_CC_pivot
```

```
##           CreditCard
## PersonalLoan  0    1
##           0 1930  792
##           1  187   91
```

Compute the following quantities

1. $P(CC = 1 \mid Loan = 1)$ (the proportion of credit card holders among the loan acceptors)

```
CCholders_loan_accepted <- loan_CC_pivot["1","1"] / sum(loan_CC_pivot["1", ])
CCholders_loan_accepted
```

```
## [1] 0.3273381
```

2. $P(Online = 1 \mid Loan = 1)$

```
onlineuser_loan_accepted <- loan_online_pivot["1","1"] / sum(loan_online_pivot["1", ])
onlineuser_loan_accepted
```

```
## [1] 0.6438849
```

3. $P(Loan = 1)$ (the proportion of loan acceptors)

```
loan_accepted <- sum(training$Personal.Loan == 1) / nrow(training)
loan_accepted
```

```
## [1] 0.09266667
```

4. $P(CC = 1 \mid Loan = 0)$

```
Cardholder_loanrejected <- loan_CC_pivot["0","1"] / sum(loan_CC_pivot["0", ])
Cardholder_loanrejected
```

```
## [1] 0.2909625
```

5. $P(Online = 1 \mid Loan = 0)$

```
inactiveuser_loanrejected <- loan_online_pivot["0","1"] / sum(loan_online_pivot["0", ])
inactiveuser_loanrejected
```

```
## [1] 0.5951506
```

6. $P(\text{Loan} = 0)$

```
loan_rejected <- sum(training$Personal.Loan == 0) / nrow(training)
loan_rejected
```

```
## [1] 0.9073333
```

Naive Bayes probability $P(\text{Loan} = 1 \mid \text{CC} = 1, \text{Online} = 1)$

```
CCholder_onlineuser <- nrow(training[training$CreditCard == 1 & training$Online == 1, ]) / nrow(training)
Loanaccepted_CCholder_onlineuser <- (CCholders_loan_accepted*onlineuser_loan_accepted*loan_accepted)/CCholder_loan_accepted
Loanaccepted_CCholder_onlineuser
```

```
## [1] 0.1101382
```

Comparing the pivot table value with the Naive Bayes Probability

```
probability_loanaccepted
```

```
## [1] 0.1071429
```

```
Loanaccepted_CCholder_onlineuser
```

```
## [1] 0.1101382
```

There is a slight differences between the two values. The Pivot table value looks more accurate as it is without any assumptions and Naive Bayes assumes independence between predictors.

Naive Bayes model

```
model <- naiveBayes(Personal.Loan ~ CreditCard + Online, data = training)
predict_probability <- predict(model, training, type = "raw")
NaiveBayes_probability <- predict_probability[training$CreditCard == 1 & training$Online == 1, 2]
NaiveBayes_probability
```

```
## [1] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [8] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [15] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [22] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [29] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [36] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [43] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [50] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [57] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [64] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [71] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [78] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [85] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [92] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [99] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
```

[illegible]

```
## [484] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [491] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [498] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [505] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [512] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [519] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
## [526] 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935 0.1156935
```

Analysis: Naive Bayes probability is more accurate than the probability form Section E because model-derived probability considers distribution and relationship within the data.